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Calculation of transmission tunneling current across arbitrary potential barriers

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This paper presents a simple method for accurately calculating quantum mechanical transmission probability and current across arbitrary potential barriers by using the multistep potential approximation. This method is applicable to various potential barriers and wells, including continuous variations of potential energy and electron effective mass. Various potential barrier structures and a hot-electron transistor are analyzed to show the feasibility of this method.

I. INTRODUCTION

Recently, from the viewpoint of high-speed and new functional device application,¹⁻³ there has been an increasing interest in resonant tunneling in quantum-well and superlattice structures. The WKB (Wentzel-Kramers-Brillouin) approximation, the conventional method used to calculate the transmission coefficient across potential barriers, fails to explain the resonance phenomena. Furthermore, the WKB method is inaccurate in regions where the potential profile varies abruptly.⁴ Abruptly varying potential functions are, however, frequently encountered at the interface between two different materials in heterojunction structures.

Another method for calculating the transmission coefficient is to solve Schrödinger's equations through potential barriers. Chandra and Eastman⁵ calculated the transmission coefficient for a triangular barrier by solving Schrödinger's equations via the numerical method. On the other hand, Gundlach⁶ calculated the tunneling current for a trapezoidal barrier by connecting the Airy functions, exact solutions for Schrödinger's equations, at two interfaces. The same procedure has been applied to triangular barriers by Christodoulides *et al.*⁷ Lui and Fukuma⁸ showed this calculation to be applicable to use with arbitrary piecewise linear potential barriers. These calculations are, however, unsuitable for designing quantum-well and superlattice structures, because of the complicated treatment involved. Furthermore, variations of electron effective masses have never been taken into account in these analyses.

This paper presents a simpler method for accurately calculating the transmission coefficient and current across arbitrary potential barriers. In the present method, we approximate variations of potential energy and electron effective mass by multistep functions (multistep potential approximation). The transmission coefficient is calculated by connecting momentum eigenfunctions.⁹ The details of the calculation procedures are described in Sec. II. As mentioned above, various potential barriers, including continuous variations of potential and effective mass, can be analyzed easily by using the present method. For example, rectangular and parabolic potential barriers, fabricated with GaAs/AlGaAs heterostructures, are analyzed in Sec. III. Section IV de-

scribes the analysis for hot-electron transistors (HETs) as an application to quantum size effect devices.

II. CALCULATION PROCEDURE

A. Transmission probability across arbitrary potential barriers

In the present calculation, instead of dealing with continuous variations of potential energy, we split the potential barrier up into segments, in which potential energy can be regarded as a constant. In the limit as the divisions become finer and finer, a continuous variation will be recovered.¹⁰

Let us assume the potential barrier to be a sequence of N small segments. An example, in the case where $N = 10$, is shown in Fig. 1, where the potential barrier $U(x)$, the effective mass $m^*(x)$, and the permittivity $\epsilon(x)$ are approximated by the multistep functions

$$U(x) = U_j = U[(x_{j-1} + x_j)/2], \quad (1a)$$

$$m^*(x) = m_j^* = m^*[(x_{j-1} + x_j)/2], \quad (1b)$$

$$\epsilon(x) = \epsilon_j = \epsilon[(x_{j-1} + x_j)/2], \quad (1c)$$

$$\text{for } x_{j-1} < x < x_j \quad (j = 0, 1, 2, \dots, N+1).$$

The wave function ψ_j in the j th region, associated with

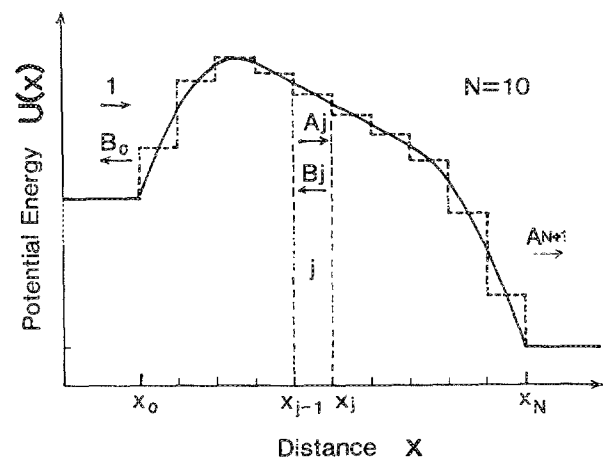


FIG. 1. Energy band diagram (solid line) and approximated potential function (broken line) for the potential barrier.

an electron with energy E moving normally on the barrier, is given by

$$\psi_j(x) = A_j \exp(ik_j x) + B_j \exp(-ik_j x), \quad (2)$$

where

$$k_j = \sqrt{[2m_j^*(E - U_j)]/\hbar}, \quad (3)$$

and $\hbar$ is the reduced Planck's constant.

$$M_l = \frac{1}{2} \begin{bmatrix} (1 + S_l) \exp[-i(k_{l+1} - k_l)x_l] & (1 - S_l) \exp[-i(k_{l+1} + k_l)x_l] \\ (1 - S_l) \exp[i(k_{l+1} + k_l)x_l] & (1 + S_l) \exp[i(k_{l+1} - k_l)x_l] \end{bmatrix} \quad (5)$$

and

$$S_l = \frac{m_{l+1}^*}{m_l^*} \frac{k_l}{k_{l+1}}. \quad (6)$$

By setting $A_0 = 1$ and $B_{N+1} = 0$ in Eq. (4) for $j = N + 1$, we can calculate the transmission amplitude A_{N+1} and the transmission probability $D(E)$ as follows:

$$A_{N+1} = \frac{m_{N+1}^*}{m_0^*} \frac{k_0}{k_{N+1}} \frac{1}{M_{22}} \quad (7)$$

and

$$D(E) = \frac{m_0^*}{m_{N+1}^*} \frac{k_{N+1}}{k_0} |A_{N+1}|^2, \quad (8)$$

where

$$M = \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix} = \prod_{l=0}^N M_l. \quad (9)$$

B. Transmission current calculation

The band diagram, used for calculating the transmission current-voltage (I - V) characteristics for the potential barrier, is shown in Fig. 2. The solid line denotes the potential function for the barrier, whereas the broken line denotes the approximated potential function. As shown in Fig. 2, the

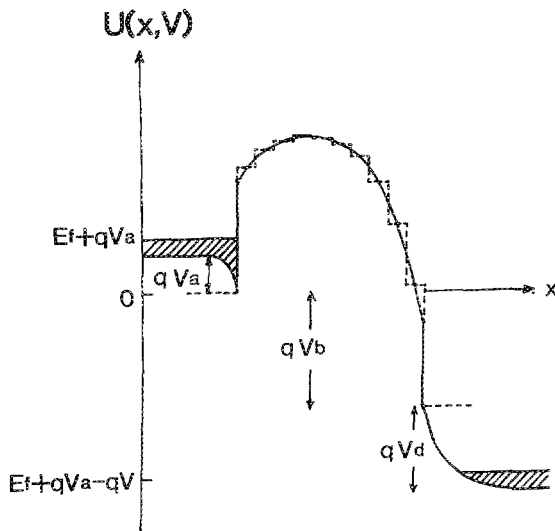


FIG. 2. Energy band diagram for the potential barrier under biasing conditions.

From the continuity of $\psi_j(x)$ and $(1/m_j^*) (d\psi_j/dx)$ at each boundary,¹¹ the determining A_j and B_j in Eqs. (2) can be reduced to the multiplication of the following $N + 1$ (2×2) matrices:

$$\begin{pmatrix} A_j \\ B_j \end{pmatrix} = \prod_{l=0}^{j-1} M_l \begin{pmatrix} A_0 \\ B_0 \end{pmatrix}, \quad (4)$$

where

total applied voltage V is expressed as $V = V_a + V_b + V_d$, where V_a , V_b , and V_d are the voltage drop values in the accumulation layer, the barrier, and the depletion layer, respectively. These values and the space charge n_s per unit area in the depletion layer, which is equal to the net charge in the accumulation layer, are determined using following equations¹²:

$$\exp(qV_a/kT) - qV_a/kT - 1 = q^2 n_s^2 / 2\epsilon_0 k T N_D, \quad (10)$$

$$V_b = \int_0^{L_b} [qn_s/\epsilon(x)] dx, \quad (11)$$

$$V_d = qn_s^2 / 2\epsilon_{N+1} N_D, \quad (12)$$

where N_D is the donor concentration in the semiconductor at both contacts, L_b is the barrier thickness, q is the electronic charge, k is the Boltzmann's constant, and T is the temperature. The Boltzmann distribution and the depletion-layer approximation are assumed for Eqs. (10) and (12), respectively.

The potential function $U(x)$ in the barrier is determined by the superimposition of the zero-bias potential and potential drop $\phi(x) = \int_0^x [qn_s/\epsilon(x)] dx$. The transmission probability $D(E_x, V)$ through the barrier region is calculated as described above. The accumulation and depletion regions are assumed not to affect the transmission probability. Assuming the dependence of transmission probability only on longitudinal electron energy E_x for oblique incidence, the current density is given by¹³

$$J(E_x) dE_x = \frac{qm_0^*}{2\pi^2 \hbar^3} D(E_x, V_b) \times \int_{E_x}^{\infty} [f_0(E) - f_{N+1}(E)] dE dE_x, \quad (13a)$$

where f_0 and f_{N+1} are the distribution functions in the left contact and in the right contact, respectively. Assuming the Fermi-Dirac distribution, Eq. (13a) can be rewritten as follows:

$$J(E_x) = \frac{qm_0^* k T}{2\pi^2 \hbar^3} D(E_x, V_b) \times \ln \left(\frac{1 + \exp(E_f + qV_a - E_x)/kT}{1 + \exp(E_f + qV_a - qV - E_x)/kT} \right). \quad (13b)$$

Here, the Fermi level in the accumulation region is treated as to be raised by qV_a .

III. EXAMPLES

Various potential barriers fabricated with GaAs/ $\text{Al}_x\text{Ga}_{1-x}\text{As}$ structures were analyzed.

The conduction-band offset ΔE_c was taken to be 60% of GaAs and $\text{Al}_x\text{Ga}_{1-x}\text{As}$ Γ band-gap difference.¹⁴ In the following, the electron concentration in GaAs was taken to be $n = 1 \times 10^{18} \text{ cm}^{-3}$, equal to N_D , and the Fermi energy was presumed to be 0.05 eV at 77 K. The parameters used are listed below¹⁵:

$$\Delta E_c = \begin{cases} 0.75x \text{ (eV)} & \text{for } x < 0.45 \\ 0.75x + 0.69(x - 0.45)^2 \text{ (eV)} & \text{for } x > 0.45, \end{cases} \quad (14a)$$

$$m^*/m_0 = 0.067 + 0.083x, \quad (14b)$$

$$\epsilon/\epsilon_0 = 13.1(1 - x) + 10.1x, \quad (14c)$$

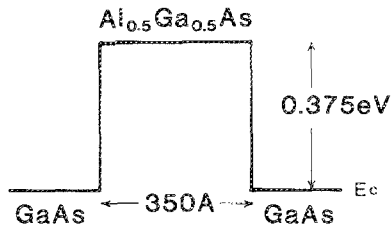
where m_0 and ϵ_0 are the free electron mass and the vacuum static dielectric constant, respectively.

A. Transmission probability across quantum barrier structures

The transmission probability is calculated in semiconductor-insulator-semiconductor (SIS) structures with a rectangular barrier and with a parabolic barrier as shown in Figs. 3(a) and 3(b), respectively.

Figure 4(a) shows the transmission probability D for $E_x = 0.05 \text{ eV}$ across the rectangular barrier [in Fig. 3(a)] with respect to V_b . In these calculations, the barrier is divided into N segments with N values ranging between 10 and 80. There is only a slight difference in solutions for $N \geq 40$ and they converge to the solid curve [in Fig. 4(a)] for $N \geq 80$. For this case, the Airy functions can give the exact solution,⁶ which coincides with the solid curve. The oscillatory behavior of $D(V_b)$ is presumed to be due to resonance through the virtual states above the barrier.¹⁶ Electron wavelengths at the resonant states are about 80 Å for $V_b = 1 \text{ V}$ and 50 Å for

(a) Rectangular Barrier



(b) Parabolic Barrier

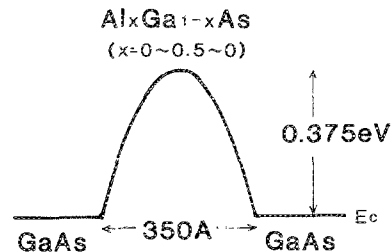


FIG. 3. Analyzed single-barrier structure band diagrams. (a) Rectangular barrier. (b) Parabolic barrier.

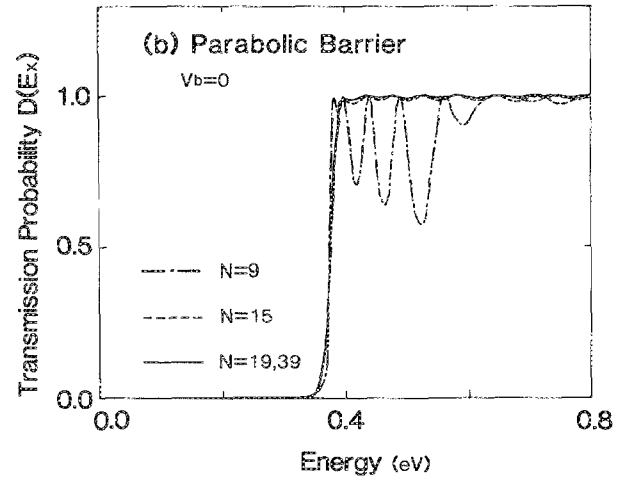
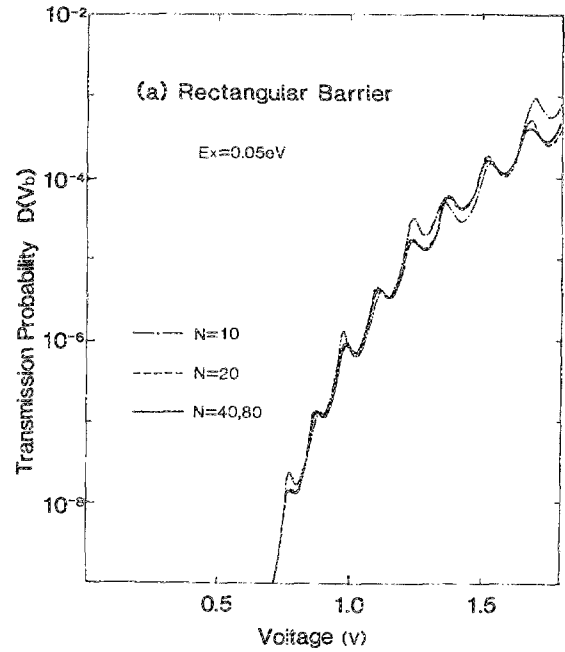


FIG. 4. Transmission probability plot (a) vs voltage for the rectangular barrier shown in Fig. 3(a), where N values range from 10 to 80, and (b) vs electron energy calculated for the parabolic barrier shown in Fig. 3(b), where N values range from 9 to 39.

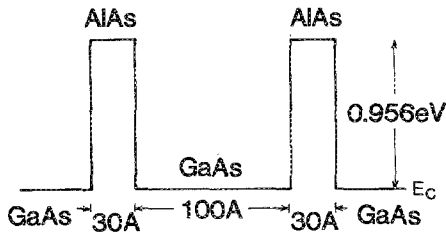
$V_b = 2 \text{ V}$, whereas the width of each segment is about 9 Å for $N = 40$. Thus, in the present method, the exact solution can be obtained by choosing a segment width sufficiently smaller than the electron wavelengths at the resonant states.

Figure 4(b) shows transmission probability D across the parabolic barrier [in Fig. 3(b)] at thermal equilibrium with respect to E_x for N values ranging between 9 and 39. The flat structure for the transmission probability profiles is a notable difference from the structure for a rectangular barrier.

B. I - V characteristics for double-barrier structures

The transmission probability and the I - V characteristics are calculated for double-barrier structures with a rectangular well and with a parabolic well, as shown in Figs. 5(a) and 5(b), respectively.

(a) Rectangular Well



(b) Parabolic Well

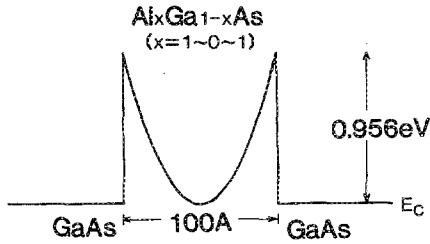


FIG. 5. Analyzed double-barrier structure band diagrams. (a) Rectangular well. (b) Parabolic well.

Figures 6(a) and 6(b) show $D(E_x)$ for a rectangular well [in Fig. 5(a)] with $N = 32$ and for a parabolic well [in Fig. 5(b)] with $N = 41$, respectively. The peaks of $D(E_x)$, E_n ($n = 0, 1, 2, \dots$), separate at regular intervals for a parabolic well¹⁷ as shown in Fig. 6(b), in contrast to that for a rectangular well shown in Fig. 6(a). These results agree well with the concept that boundary state energies are expressed as $E_n = (n + 1/2)\hbar\omega$ for the quantum well expressed as $U(x) = (1/2)m^*\omega^2x^2$.

Calculated I - V characteristics for both structures at 77 K are shown in Figs. 7(a) and 7(b). Figures 7(a) and 7(b) correspond to the rectangular well and the parabolic well, respectively. In these figures, the solid lines denote the results, including the effects of accumulating and depleting, whereas the broken lines denote the results without these effects. With the accumulation and depletion taken into account, the current density increases and the voltage shifts. These results show that voltage drops at contact layers seriously affect I - V characteristics, and hence, should be taken into account in analyzing and designing quantum size devices.

IV. APPLICATION—ANALYSIS OF HETs (HOT-ELECTRON TRANSISTORS)

The HET, one of the quantum size effect devices utilizing electron tunneling through potential barriers, was analyzed to show the feasibility of this method.

A. Analysis procedure for HETs

Figure 8 is a band diagram of the HET proposed by Heiblum.¹⁸ The I - V characteristics of HETs can be analyzed applying the present calculation to the potential function $U(x, V_{BE}, V_{CB})$ (shown in Fig. 8), where V_{BE} is the voltage applied between emitter and base, and V_{CB} is the voltage between the base and collector. The present calculation is the

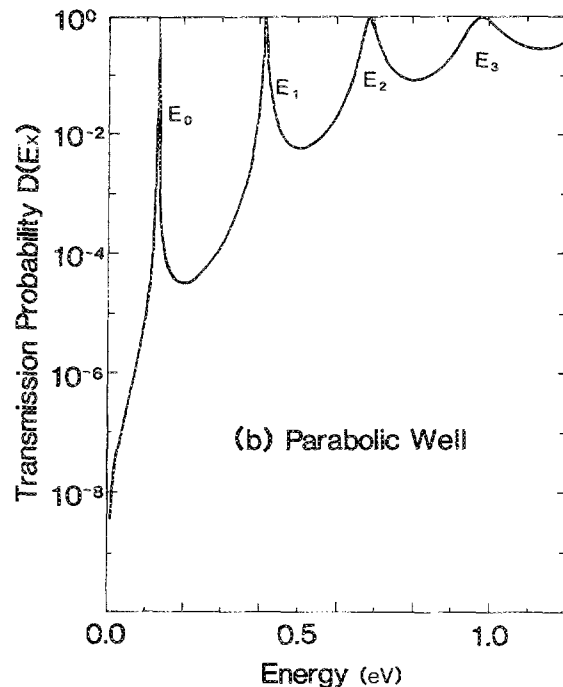
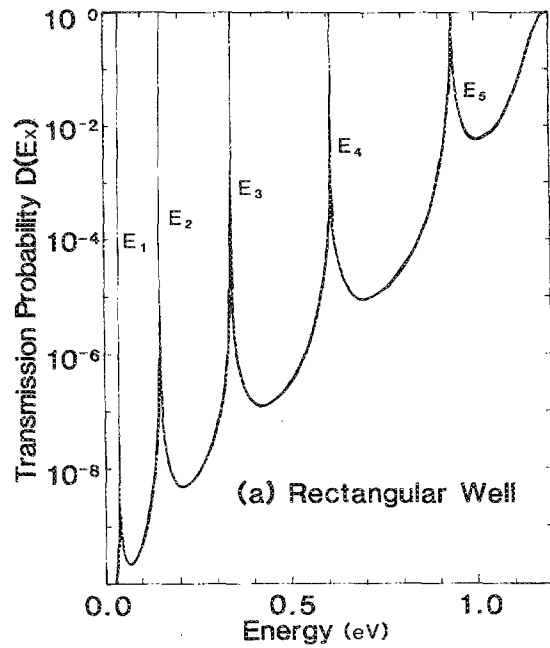


FIG. 6. Transmission probability vs electron energy plot (a) calculated for the rectangular well structure shown in Fig. 5(a) and (b) calculated for the parabolic well structure shown in Fig. 5(b).

extension of that used for MOMOM (metal-oxide-metal-oxide-metal) devices.¹⁸

The transmission probability $D_E(E_x, V_{BE})$ across the emitter barrier and $D_C(E_x, V_{CB})$ across the collector barrier can be calculated, as described in Sec. II A. Current density J_E , between the emitter and base, was calculated as described in Sec. II B. That is,

$$J_E(E_x) = \frac{qm^*}{2\pi^2\hbar^3} D_E(E_x) \int_{E_x}^{\infty} [f_E(E) - f_B(E)] dE, \quad (15)$$

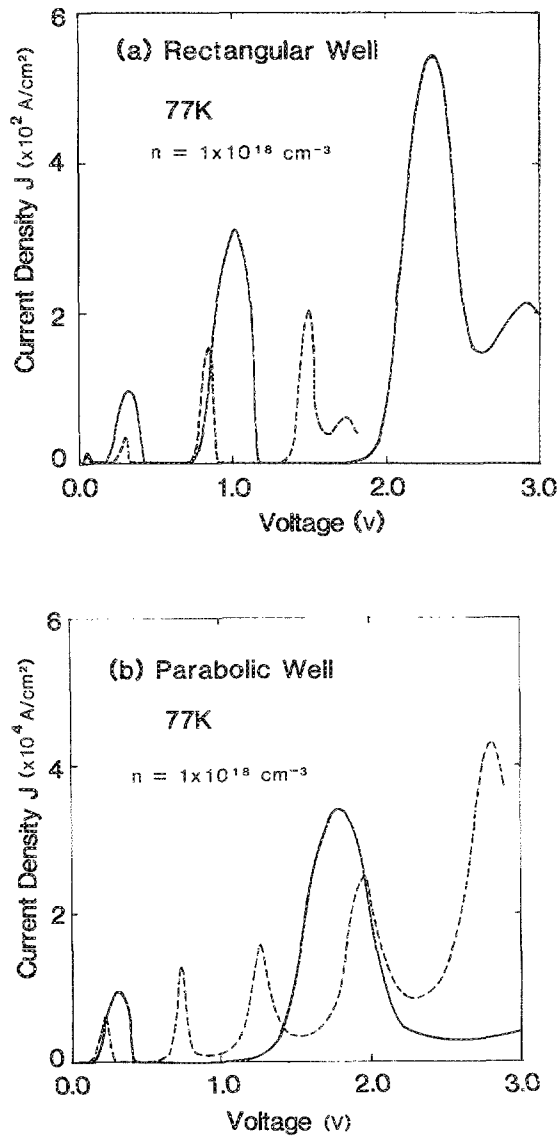


FIG. 7. Current density vs applied voltage plot at 77 K, including the accumulation and depletion (solid line) and without these effects (broken line), (a) calculated for the rectangular well structure shown in Figs. 5(a) and (b) calculated for the parabolic well structure shown in Fig. 5(b).

where $f_E(E)$ and $f_B(E)$ are the Fermi-Dirac distribution functions in the emitter and in the base, respectively. Current density J_C into the collector is expressed as $J_C = J_C^* + J_{CB0}$, where J_C^* is the current into the collector due to

$$\frac{1}{\tau_B} = \begin{cases} 0 & \text{for } E'_x < 0.05 \text{ eV} \\ \frac{1}{3} \times 10^{14} (E'_x - 0.05) \text{ (s}^{-1}\text{)} & \text{for } 0.05 \text{ eV} < E'_x < 0.2 \text{ eV} \\ 2 \times 10^{13} \text{ (s}^{-1}\text{)} & \text{for } E'_x > 0.2 \text{ eV,} \end{cases} \quad (18)$$

where

$$E'_x = E_x + qV_{BE}.$$

B. Calculated transfer characteristics for a HET

Transfer characteristics were calculated for the HET (shown in Fig. 8) operating under common emitter conditions.

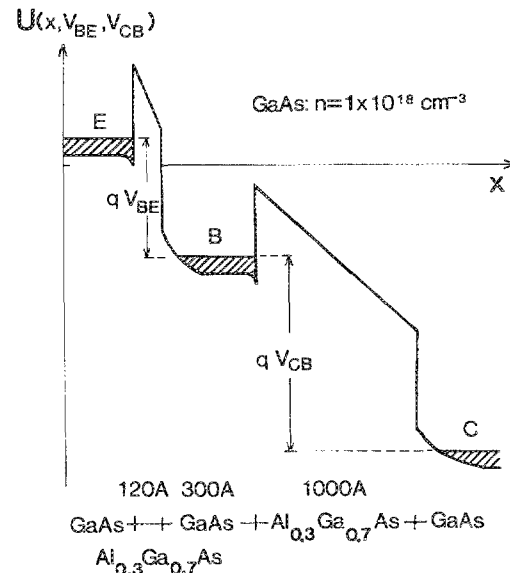


FIG. 8. Energy band diagram for the HET.

injection through the emitter, and J_{CB0} is the tunneling current between the collector and base when the emitter is not connected. J_{CB0} can be calculated similarly to J_E , using substitutions D_C for D_E , f_B for f_E , and f_C for f_B in Eq. (15). J_C^* can be calculated as follows:

$$J_C^*(E_x) = \frac{qm^*}{2\pi^2\hbar^3} D_E(E_x) T_B(E_x) D_C(E_x) \times \int_{E_x}^{\infty} f_E(E) [1 - f_B(E)] [1 - f_C(E)] dE \quad (16)$$

and

$$T_B(E_x) = \exp[-L_B/v_B(E_x)\tau_B(E_x)], \quad (17)$$

where L_B is the thickness in the base, $v_B(E_x)$ the velocity for the x direction of the hot electrons in the base, and $\tau_B(E_x)^{-1}$ the scattering ratio of hot electrons in the base.

Levi *et al.* calculated the dependence of the scattering ratio of hot electrons in GaAs on the injection energy including the coupled plasmon-phonon mode.¹⁹ We chose the following function to approximate their result for $n = 1 \times 10^{18} \text{ cm}^{-3}$:

The calculated transfer ratio $\alpha = J_C/J_E$ and differential transfer ratio $\alpha' = dJ_C^*/dJ_E$ are shown in Figs. 9 and 10, respectively. The effects of accumulating and depleting are also taken into account. α' peaks at $V_{BE} = 0.2 \text{ V}$, following an initial rapid increase, and rises again slowly with V_{BE} increasing above 0.25 V. The rapid increase results from the abrupt increase in the collection efficient $D_C(E_x)$, as the injection energy for hot electrons is raised above the collec-

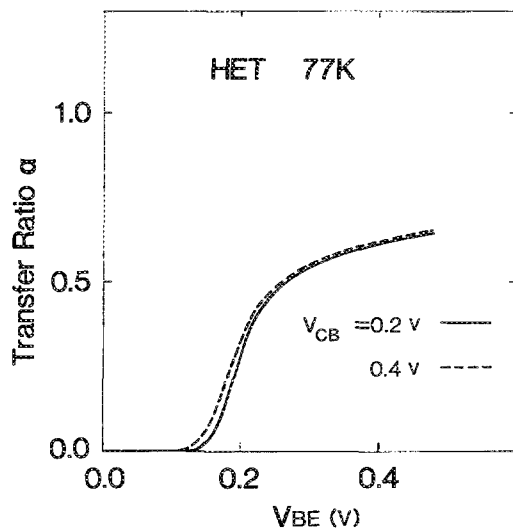


FIG. 9. Transfer ratio vs base-emitter voltage plot at 77 K. Calculated for the HET (shown in Fig. 8) under common emitter biasing conditions.

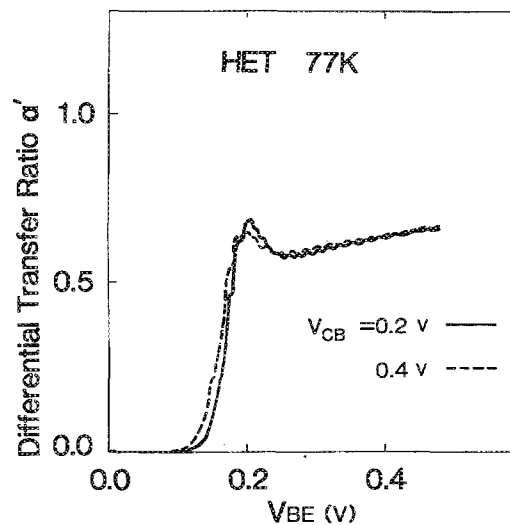


FIG. 10. Differential transfer ratio plot vs base-emitter voltage plot at 77 K. Calculated for the HET (shown in Fig. 8) under common emitter biasing conditions.

tor barrier. The slow increase is related to the gradual increase in the mean free path of hot electrons, as the scattering rate saturates. The periodic structure in α' , also observed in experiments,²⁰ is due to the quantum mechanical reflection at the collector barrier. However, in this calculation, the quantum mechanical reflection through the quantum well, formed by the base layer, is not taken into account.

V. CONCLUSION

A simple method for accurately calculating the transmission current across arbitrary potential barriers is presented. By using this method, various potential barriers and wells can be analyzed, including continuous variations of potential and effective mass. Therefore, the present method is applicable to heterojunction structures such as superlattice structures. Furthermore, the present method is shown to be useful for analyzing and designing the quantum size effect devices.

Although SIS barriers are analyzed in this paper, of course, this calculation can be extended to the analysis of the metal-semiconductor barriers, the metal-insulator-semiconductor barriers, and so on. Further results will be reported elsewhere.

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